

NOVEXA TECHNOLOGIES

SALES PERFORMANCE & BUSINESS INTELLIGENCE DASHBOARD

Data Analysis Internship Project

SUPERSTORE SALES DATA | 2015–2018

9,800 Records	\$2.26M Total Sales	4,922 Orders	793 Customers
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1. Executive Snapshot

This project transforms Superstore transaction data into a practical business intelligence view of sales performance. The workflow combines Python-based exploratory data analysis with an interactive Power BI dashboard to examine sales across time, categories, regions, customer segments, products, and shipping modes.

The final dashboard is designed for fast decision support: headline KPIs provide an immediate business snapshot, while interactive slicers allow users to explore the same measures by year, region, category, and segment.

2. Project Objective

The objective is to analyze historical sales transactions and build an interactive business intelligence dashboard that highlights sales patterns and operational indicators. The analysis is descriptive; profitability is outside scope because the supplied dataset does not contain a Profit field.

- Measure overall sales, orders, customers, and average shipping time.
- Identify sales patterns across years and months.
- Compare sales performance by category, region, and customer segment.
- Identify the top products by sales value.
- Examine sales distribution across shipping modes.
- Present the results through an interactive Power BI dashboard.

3. Dataset Overview

The analysis uses the Superstore sales dataset supplied as `superstore.csv`. The dataset contains 9,800 transaction records and 18 columns covering order information, customer details, product information, geography, shipping information, and sales.

Dataset characteristic	Result	Notes
Rows	9,800	Transaction-level records
Columns	18	Sales, customer, product, shipping and location fields
Date range	2015–2018	03/01/2015 to 30/12/2018
Missing values	11	All 11 occur in Postal Code
Duplicate rows	0	No duplicate records identified
Unique orders	4,922	Distinct Order IDs
Unique customers	793	Distinct Customer IDs

4. Tools & Technologies

- Python — exploratory data analysis and descriptive statistics.
- Pandas — data loading, cleaning, transformation, grouping and aggregation.
- Power BI Desktop — interactive dashboard creation and business intelligence reporting.
- DAX — KPI measures and the Shipping Days calculated column.
- Microsoft Word — project documentation and final report.

5. Data Preparation & Quality Checks

The dataset was loaded into Python and checked before analysis. Date fields were converted to proper datetime values, while missing-value and duplicate checks were performed. The analysis identified 11 missing Postal Code values and no duplicate rows. Because Postal Code is not required for the core sales KPIs and visual analysis, no sales values were affected by the missing entries.

A Shipping Days field was derived from the difference between Ship Date and Order Date. The same business logic was implemented in Power BI using a DAX calculated column so that the dashboard could report average shipping time interactively.

DAX logic: `Shipping Days = DATEDIFF(Order Date, Ship Date, DAY)`

6. Exploratory Data Analysis

6.1 Sales Overview

Total sales across the dataset are \$2,261,536.78. Individual transaction sales have a mean of \$230.77 and a median of \$54.49, indicating that the distribution contains a relatively small number of high-value transactions compared with the typical transaction.

6.2 Sales by Category

Category	Sales
Technology	\$827,455.87
Furniture	\$728,658.58
Office Supplies	\$705,422.33

6.3 Sales by Region

Region	Sales
West	\$710,219.68
East	\$669,518.73
Central	\$492,646.91
South	\$389,151.46

6.4 Sales Trend

Annual sales were \$479,856.21 in 2015, \$459,436.01 in 2016, \$600,192.55 in 2017, and \$722,052.02 in 2018. The monthly analysis covers all 48 months; the highest monthly sales value occurred in November 2018 at \$117,938.16, while the lowest occurred in February 2015 at \$4,519.89.

6.5 Customer Segments

Consumer generated \$1,148,061.12 in sales, followed by Corporate at \$688,494.91 and Home Office at \$424,980.75.

6.6 Top Products

The highest-sales product in the analysis was the Canon imageCLASS 2200 Advanced Copier with \$61,599.82 in sales. Other high-sales products include the Fellowes PB500 Electric Punch Plastic Comb Binding Machine (\$27,453.38) and the Cisco TelePresence System EX90 Videoconferencing Unit (\$22,638.48).

6.7 Shipping Performance

Average shipping time across the dataset is 3.96 days. Average shipping time by ship mode was approximately 0.04 days for Same Day, 2.18 days for First Class, 3.25 days for Second Class, and 5.01 days for Standard Class.

7. Key Business Insights

Sales concentration — Technology generated the highest category sales at approximately \$827.46K.

Geographic pattern — The West region recorded the highest sales at approximately \$710.22K.

Growth pattern — 2018 recorded the highest annual sales in the dataset at approximately \$722.05K.

Customer mix — The Consumer segment contributed approximately \$1.15M in sales.

Product concentration — A small group of products accounts for comparatively high sales values, making the Top 10 Products view useful for focused product analysis.

Shipping view — Average shipping time is 3.96 days overall, with substantial variation by ship mode.

8. Power BI Dashboard

The final Power BI dashboard combines four headline KPI cards with interactive analytical visuals. The KPI cards display Total Sales, Total Orders, Total Customers, and Average Shipping Days. Supporting visuals show the sales trend over time, sales by category, sales by region, sales by segment, Top 10 Products, and sales by Ship Mode.

Four slicers — Year, Region, Category, and Segment — allow users to filter the dashboard dynamically. When slicer selections are cleared, the dashboard returns to the full-dataset view.

[INSERT POWER BI DASHBOARD SCREENSHOT HERE]

9. Technical Implementation

Python EDA Workflow

- Load the CSV file with Pandas.
- Parse Order Date and Ship Date as datetime fields.
- Inspect shape, data types, missing values and duplicates.
- Calculate descriptive statistics and total sales.
- Aggregate sales by category, sub-category, region, year, month and segment.
- Identify top customers and top products by sales.
- Calculate Shipping Days and shipping performance by Ship Mode.
- Calculate unique orders, unique customers and average sales per order.

Power BI Implementation

- Imported superstore.csv into Power BI Desktop.
- Verified date and numeric data types.
- Created DAX measures for Total Sales, Total Orders and Total Customers.
- Created the Shipping Days calculated column using DATEDIFF.
- Created the Avg Shipping Days measure.

- Built interactive KPI cards, charts and slicers.
- Saved the final report as Sales_Performance_Dashboard.pbix.

10. Limitations & Scope

- The dataset does not contain a Profit field, so profitability and profit-margin analysis were outside the scope of this project.
- The analysis is descriptive and historical; it does not forecast future sales.
- Missing Postal Code values were present in 11 records.
- The findings describe the supplied dataset and period (2015–2018) and should not automatically be treated as current business performance.

11. Conclusion

The project demonstrates an end-to-end data analysis and business intelligence workflow: raw transaction data was validated and explored in Python, meaningful sales metrics were calculated, and the results were translated into an interactive Power BI dashboard. The final solution provides a concise view of sales performance across time, geography, categories, customer segments, products, and shipping modes.

12. Project Deliverables

Deliverable	File
Dataset	superstore.csv
Python analysis	EDA_script.py
Power BI dashboard	Sales_Performance_Dashboard.pbix
Documentation	Sales_Performance_Business_Intelligence_Report.docx